

A Robust Position Estimation Algorithm under Unusual Large Range Errors

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Abstract— This paper provides a robust fusion algorithm for accurate position estimation under uncertain large errors in range measurements. Many researchers have studied UWB, IMU, odometry integration algorithms for estimating position, velocity, attitude and IMU biases. However, the fused results of these conventional algorithms are likely to be affected by the unusual ranging measurement errors. Therefore, in order to improve the positioning accuracy under the large errors of the UWB range, a robust fusion algorithm is required. In this paper, instead of using the range directly, the estimated range that the errors is reduced is used. Reducing the errors is possible by using the odometry velocity that is relatively accurate and is independent of range measurement. The robustness and accuracy of the proposed algorithm is verified by a mobile robot with real-time positioning and trajectory control under large range errors which are occurred randomly.

I. INTRODUCTION

Although many users are benefiting from the development of the global navigation satellite system (GNSS), there are still many situations where the GNSS cannot operate, such as indoor environment, space and various GNSS-denied. For these cases, various methods such as radio frequency identification (RFID), Wi-Fi, visual sensor, inertial navigation system (INS), pseudo-lite and ultra-wideband (UWB) have been studied. However, there are still difficulties in developing a reliable system.

Among them, UWB is used in this paper to measure a range, because it provides high precision range measurements [1] with advances in many technologies such as solving multipath propagation [2]. In addition, complementary metal-oxide-semiconductor (CMOS) based UWB systems, as defined in IEEE 802.15.4a [3], are considered to be the basis of future alternative positioning systems [4]. The accuracy of UWB is experimentally measured up to the millimeter unit [5], but in practice, it has an unexpected larger error by the lack of integration, channel arbitration and clock synchronization [6]. Furthermore, signal disturbance by various surroundings and dynamic situations also increase the error [7]. The error also occurs depending on the signal strength, the radiation direction of the UWB antenna and UWB data settings for range calculation. As for the characteristics of the error, there is also a problem that it includes a bias error as well as a noise error [8]. Therefore, for more accurate and precise position calculation, it is necessary to complement each other's disadvantages by integrating with other sensors rather than positioning by using UWB alone. For this purpose, UWB, inertial measurement unit (IMU) and odometry were

integrated to improve position accuracy in various way [9], [10], [11], [12].

In this paper, the algorithm developed in reference [13] was used basically. This algorithm estimates position, velocity, heading and IMU biases as the states. In general, a sufficiently improved position result can be obtained with the algorithm of [13], but if unusual large errors are occurred in the range, the algorithm in [13] cannot provide position with satisfactory error level. In order to overcome this unexpected situation, this paper proposes a robust algorithm that the integration will not be affected even if this kind of error occurs in the range.

This paper organized as follows: section II describes and derives the proposed algorithm, section III verifies performance of the proposed algorithm with a mobile robot and UWB range sensors, and finally section IV gives the concluding remarks.

II. ODOMETRY-FUSED RANGE ESTIMATION ALGORITHM

A. Overview

Block diagrams for the conventional algorithm in [13] and the proposed system are shown in Fig 1 and Fig 2, respectively. Basically, both block diagrams are for extended Kalman filter (EKF) integration of UWB/IMU/Odometry.

As shown in Fig 1, the conventional integration algorithm uses the position calculated from the UWB range, the velocity received from the odometry, and the heading calculated from the estimated velocity as the measurement of the EKF. Acceleration and angular velocity data received from the IMU are used for INS propagation. EKF also estimates the IMU bias to remove the bias of the IMU data.

The most different part of the proposed algorithm from the conventional algorithm is the input part of the range. In Fig 1, UWB range measurement directly enters least squares calculating position, whereas in Fig 2, the estimated range is used as least squares input and UWB range measurement enters EKF for range estimation.

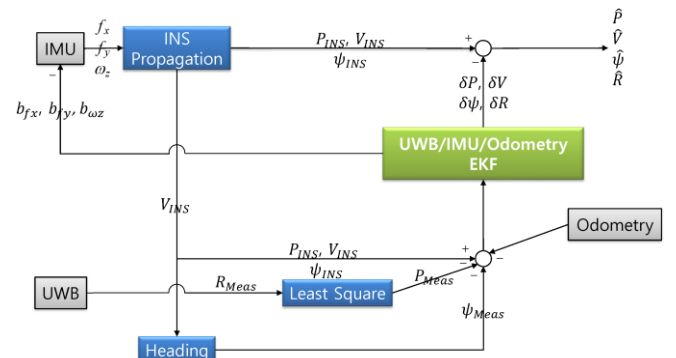


Figure 1. Block diagram of conventional algorithm

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$$\delta \dot{\mathbf{x}} = \mathbf{F} \delta \mathbf{x} + \mathbf{G} \mathbf{W} \quad (9)$$

$\delta \mathbf{x}$ is a error state vector and $\mathbf{W}$ is a process noise matrix.

$$\mathbf{F} = \begin{bmatrix} \frac{\partial \dot{P}_y}{\partial P_y} & \frac{\partial \dot{P}_y}{\partial P_x} & \frac{\partial \dot{P}_y}{\partial V_y} & \frac{\partial \dot{P}_y}{\partial V_x} & \frac{\partial \dot{P}_y}{\partial \psi} & \frac{\partial \dot{P}_y}{\partial b_{f_x}} & \frac{\partial \dot{P}_y}{\partial b_{f_y}} & \frac{\partial \dot{P}_y}{\partial b_{\phi_z}} & \frac{\partial \dot{P}_y}{\partial R_i} & \frac{\partial \dot{P}_y}{\partial R_m} & \dots & \frac{\partial \dot{P}_y}{\partial R_m} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \ddots \end{bmatrix} \quad (10)$$

$$\mathbf{G} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \vdots & \vdots & \vdots \end{bmatrix} \quad (11)$$

Note that the number of matrix elements are varied according to the number of ranges.

Measurement update proceeds as shown in Fig 3. Also, it is performed according to the frequency at which each measurement is measured. The measurement model is

$$\mathbf{z} = \mathbf{H} \mathbf{x} + \mathbf{v} \quad (12)$$

When $\mathbf{z}$ is measurement state vector, $\mathbf{H}$ is a matrix that derives the measurement from the state space and $\mathbf{v}$ is the measurement noise that is assumed to be zero mean Gaussian white noise with covariance $\mathbf{V}; \mathbf{v} \sim N(0, \mathbf{V})$ but actually UWB has no zero mean.

The measurement model is divided into three cases depending on the type of measurement input and the availability of measurement.

1) Odometry Velocity Measurement

In the first case, we update the odometry speed data by dividing it to x and y axes using the estimated heading becomes

$$\mathbf{z}_{odom} = [V_{odom,Y} \quad V_{odom,X}]^T \quad (13)$$

Odometry speed measurement is calculated from the encoder value of each left and right motor as (14).

$$V_{odom} = \frac{r_{wheel} (\omega_{left} + \omega_{right})}{2} \quad (14)$$

Where r_{wheel} is the radius of the wheel, and ω is the angular velocity of the wheels.

2) UWB Range and Position Measurement

The second case is that the UWB range and position measurement are updated and the robot moves at low speed. In this case the heading measurement is not updated because the low speed increase heading error (15).

$$\mathbf{z}_{UWB1} = [P_{meas,Y} \quad P_{meas,X} \quad R_{UWB,1} \quad \dots \quad R_{UWB,m}]^T \quad (15)$$

R_{UWB} is range measurement from UWB sensor and $\mathbf{P}_{meas}$ is calculated by the least square method (16) using the estimated range.

$$\begin{aligned} \mathbf{P}_{meas} &= (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{B} \\ \mathbf{P}_{meas} &= [P_{meas,X} \quad P_{meas,Y}]^T \end{aligned} \quad (16)$$

$\mathbf{A}$ and $\mathbf{B}$ are from (17) and (18). The matrix form of (17) is as follows (18).

$$\hat{\mathbf{e}}_{robot}^{tag i} \mathbf{P}_{meas} = \mathbf{P}^{tag i} \hat{\mathbf{e}}_{robot}^{tag i} - \hat{R}_{robot}^{tag i} \quad (i = 1, 2, \dots, m) \quad (17)$$

$\hat{R}$ is the estimated range.

$$\mathbf{A} \mathbf{P}_{meas} = \mathbf{B} \quad (18)$$

3) UWB Range, Position and Heading Measurement

The third case is when the robot moves at high speed. So the heading measurement is updated with the UWB range and position measurement (19).

$$\mathbf{z}_{UWB2} = [P_{meas,Y} \quad P_{meas,X} \quad \psi_{meas} \quad R_{UWB,1} \quad \dots \quad R_{UWB,m}]^T \quad (19)$$

The heading measurement is calculated using the estimated speed as in (20)

$$\psi_{meas} = \tan^{-1} \left(\frac{\hat{V}_x}{\hat{V}_y} \right) \quad (20)$$

$\hat{\mathbf{V}}$ is the estimated velocity.

The reason for distinguishing the second and third cases of the measurement model is that the error of the heading measurement varies depending on the speed. The heading should be calculated only if the speed is faster than appropriate speed because if the heading is calculated when the speed is too slow, the error becomes large. To obtain the speed threshold for the heading calculation, firstly set the error level of the heading you want to use as a measurement. Then, as shown in Fig 5, the velocity threshold is calculated using the relationship between velocity error and heading error [15] and it gives

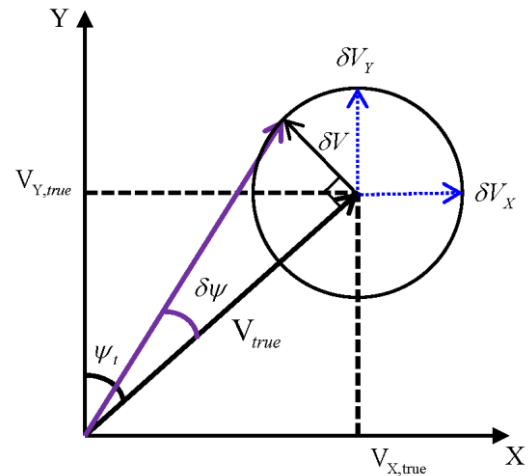


Figure 5. Relation of velocity error and heading error



Figure 6. Experiment environment

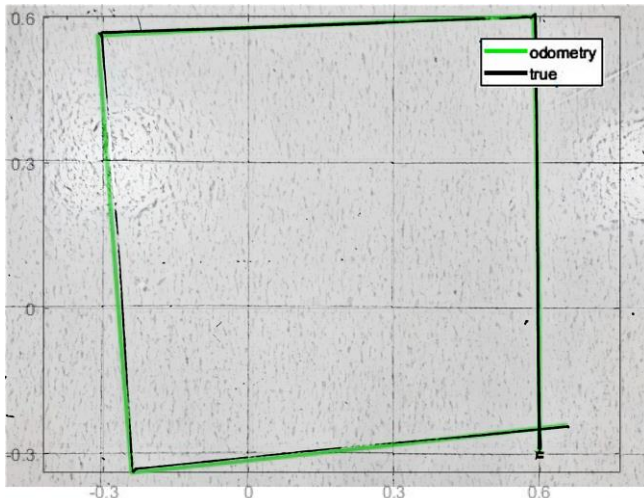


Figure 7. Comparison of drawn trajectory photographs and calculated trajectories graph

$$\begin{aligned} \delta V &\sim N(0, \sigma_v^2) & \delta \psi &\sim N(0, \sigma_\psi^2) \\ \tan \delta \psi &= \frac{\delta V}{|V_{true}|} \\ \therefore |V_{threshold}| &= \frac{\sigma_v}{\tan \sigma_\psi} \end{aligned} \quad (21)$$

Where δV is the error of speed, $\delta \psi$ is the error of heading, σ_v is the standard deviation of speed and σ_ψ is the standard deviation of heading.

If the speed of the robot is less than the threshold, the second case is applied, whereas if the speed is greater than the threshold, the third case is applied.

III. EXPERIMENT

For the experiment, a mobile robot (Turtlebot 3) equipped with Raspberry Pi, UWB sensor (Pozyx) and IMU is used.

Raspberry Pi receives commands from the server computer and reads sensor data from UWB, IMU and motor encoder.

A. Experiment configuration

Experiments were carried out in the experimental space of Fig 6. Four UWB sensors were placed at fixed positions and one UWB sensor was attached to the robot. UWB sensors give the ranges between each sensor. The space where the four sensors were placed is 2.1m x 2.7m. Because of the 2D computation, four fixed sensors were placed at the same height of the sensor mounted on the robot to equalize the height. Also, if the sensors are placed on the floor, it is difficult to measure the range stably due to the multipath error caused by reflection on the floor. Therefore, for normal experiment, the sensor should be at an appropriate height.

The robot used in the experiment is operated with two motors and is equipped with UWB sensor, IMU and Raspberry Pi. The UWB sensor, the IMU sensor and motor encoder transmit data to Raspberry Pi at 10 Hz, 100 Hz and 100 Hz respectively.

A remote computer sends an operation command 'move or 'target trajectory' to the robot. In the robot (Raspberry Pi), the real-time position calculation is continued through the EKF, and the control command is generated according to the operation command received from the remote computer and the control command is transmitted to the motor.

The experiment scenario is to compare the position error of the proposed algorithm with that of the conventional algorithm. Robot control for this scenario is distance control and heading control using only odometry and angular velocity integral instead of position control using fused position. This allows independent comparisons of true position and calculated position.

For the position error calculation, reference position data is required. In this error calculation, the position obtained by the gyroscope angular velocity integral and odometry is used as a reference. Fig 7 is the basis for whether this position data can be used as a reference. The black trajectory in Fig 7 is the true trajectory that was drawn on the floor by the robot with a pen attached on the robot's center of rotation. The green trajectory is a plot of the trajectory calculated using the angular velocity integral and odometry. Then the two trajectories were compared on a synchronized grid. As can be seen in Fig 7, the green trajectory is reasonable as a reference data, considering that the errors of the data to be compared with this reference are relatively large. Therefore, for calculating the position error, the position data by the angular velocity integral and odometry will be used as the reference. In the case of odometry, errors are accumulated as the travel distance increases and errors due to slip and sliding may occur depending on the floor environment. However, in this experiment, since the angular velocity integral is used with odometry and the travel distance is not long, the green trajectory could be close to the true (black) trajectory as shown in Fig 7.

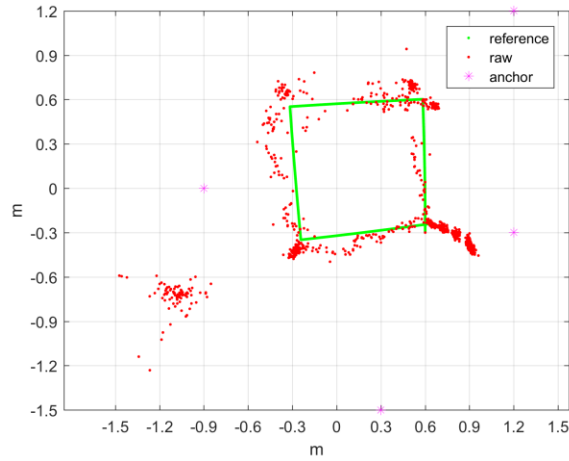


Figure 8. UWB only position result with unusually large error

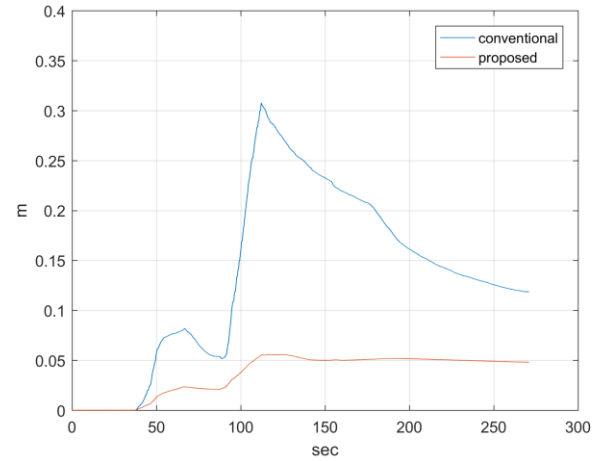


Figure 11. Position error of conventional algorithm and proposed algorithm

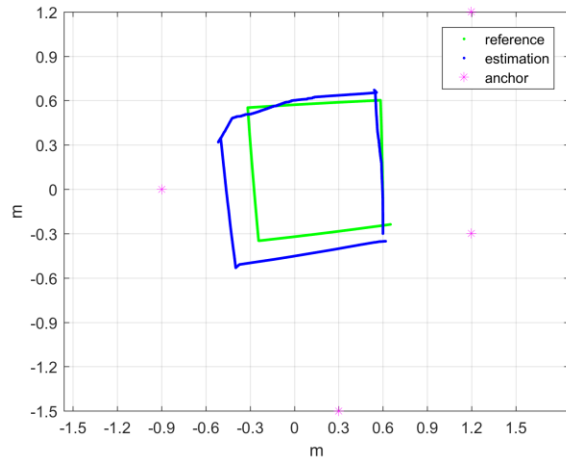


Figure 9. Integration position result using conventional algorithm

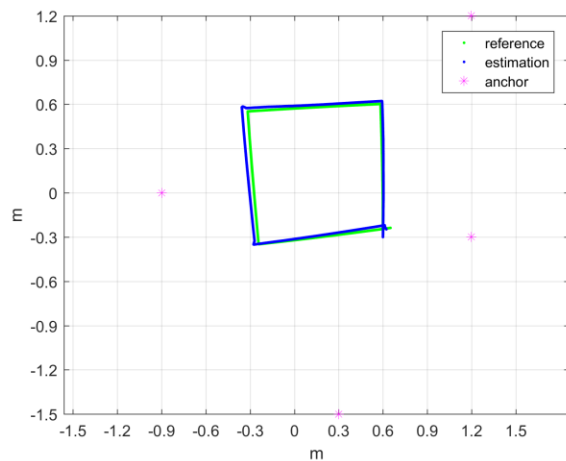


Figure 10. Integration position result using proposed algorithm



Figure 12. Capture of trajectory control video

B. Results

Fig 8 shows the position result using UWB range only when an unusually large error occurs. Points at the bottom left of the graph are far from the reference due to unusually large errors. As for this case, the integration position results using the conventional algorithm and the proposed algorithm are shown in Fig 9 and Fig 10, respectively. It can be confirmed that the error remarkably improved even if only the trajectories are compared. The graphs plotting the errors are shown in Fig 11. For this case, the RMS error of the proposed algorithm was improved about 73% over the conventional algorithm in this experiment.

The degree of improvement depends on the characteristics and size of the errors that occurs. However, the different error cases in three additional experiments showed an average of 74% RMS error improvement.

In addition, trajectory control is performed by applying this algorithm. The control trajectory travels a square trajectory with a size of 1.2x1.2 as shown in Fig 12. As a result, it is confirmed that the trajectory control of the robot with the proposed algorithm is performed well along the commanded trajectory.

IV. CONCLUSIONS

In the conventional integration algorithm, when the range error is abnormally large, it is confirmed that even if the reliability of the measurement is changed by the EKF covariance value, the error cannot improve in the conventional integration result. Therefore, we added the ranges to the state space to estimate a more accurate range. Then we set up a nonlinear equation for the range using the velocity estimated by the odometry velocity measurement that is relatively more precise and independent of the range. In this estimated range, abnormal and large errors were remarkably reduced, and the errors in the estimated position results also improved. In order to verify the actual performance, experiments were performed. The proposed algorithm showed an average improvement of about 74% over the conventional algorithm. In this paper, it is applied only to errors occurring in UWB range, but of course it is also applicable to errors that occur in all kinds of range measurement. In the future work, we will extend the proposed algorithm to methods using the UKF and/or the particle filter.

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